CHL5233: Quarto in-class assignment

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title: “CHL5233: Quarto in-class assignment” author: “Pengyu Wang” format: pdf: geometry: - top=30mm - left=20mm - heightrounded —

Setup

Read in data and create an analytical data set

First: Read countries.txt

```
countries <- read.table(
  "../data/raw/countries.txt",
  header = TRUE
)

dim(countries)
```

```
[1] 186   2
```

There are 186 countries and 2 columns in this data set.

Second: Read regions.txt

```
regions <- read.table(
  "../data/raw/regions.txt",
  header = TRUE
)

dim(regions)
```

```
[1] 249   6
```

There are 249 countries and 6 columns in this data set.

Third: Create an analytical data set

```
# Check variable names
names(countries)
```

```
[1] "country_name" "ISO"
```

```
names(regions)
```

```
[1] "country_name"    "country_code"    "region"          "sub_region"
[5] "region_code"     "sub_region_code"
```

```
# Identify the ISO country code column
possible_names <- c(
  "country_code", "iso3", "iso3c",
  "iso_code", "code", "ISO3", "ISO"
)

code_col <- intersect(possible_names, names(countries))[1]

# Standardize ISO codes
countries <- countries |>
  mutate(
    country_code = toupper(
      trimws(as.character(.data[[code_col]]))
    )
  )

regions <- regions |>
  mutate(
    country_code = toupper(
      trimws(as.character(country_code))
    )
  )

# Keep only countries included in the study
study_regions <- regions |>
```

```
  semi_join(countries, by = "country_code")

dim(study_regions)
```

```
[1] 186   6
```

The analytical data set contains 186 countries.

Tabulate countries by region and sub-region

Number of countries by region

```
region_counts <- study_regions |>
  group_by(region) |>
  summarise(
    Count = n_distinct(country_code),
    .groups = "drop"
  ) |>
  arrange(desc(Count))

kable(
  region_counts,
  col.names = c("Region", "Count"),
  caption = "Number of Countries by Region"
)
```

Table 1: Number of Countries by Region

Region	Count
Africa	54
Asia	46
Europe	40
Americas	35
Oceania	11

Number of countries by sub-region

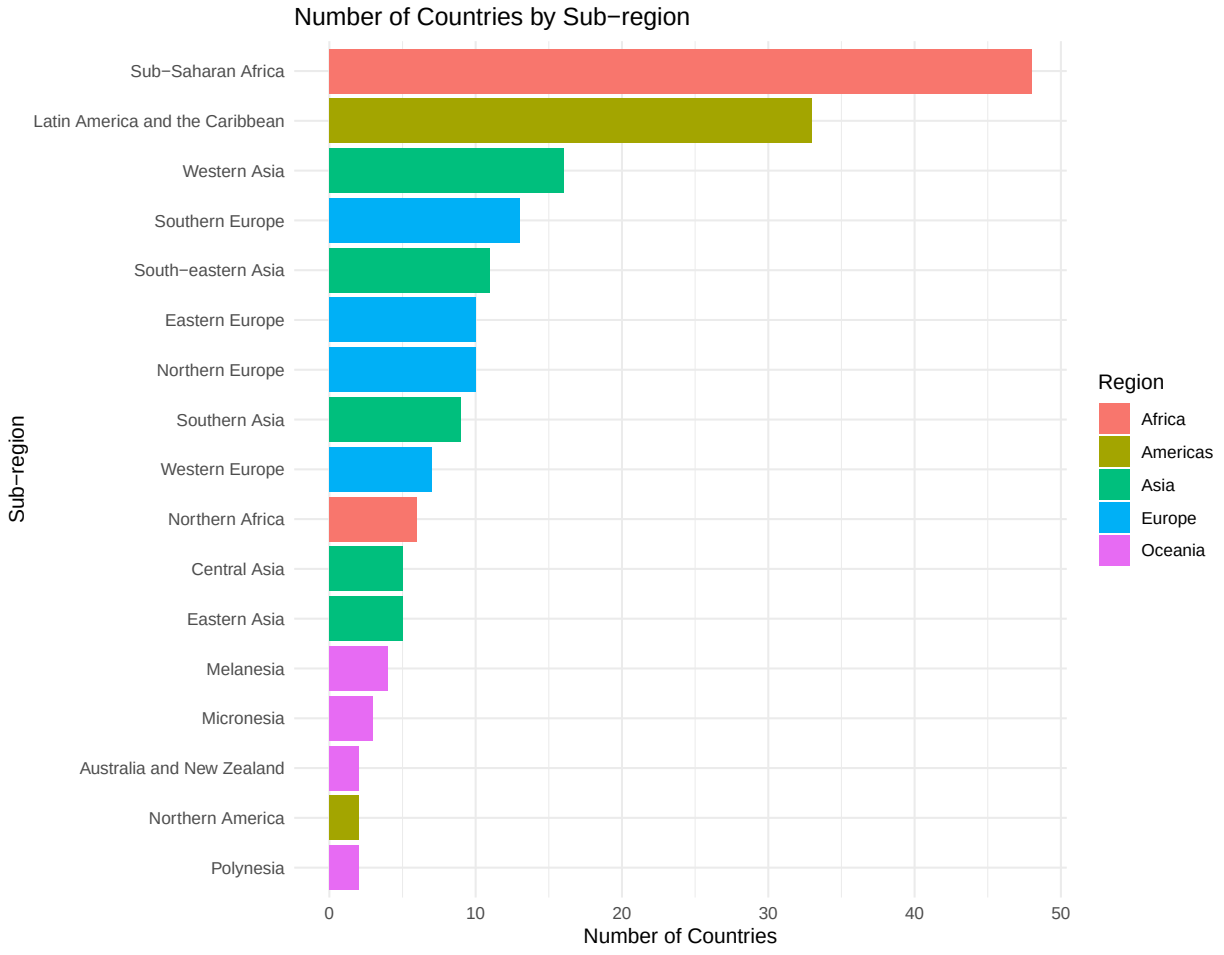
```r
sub_region_counts <- study_regions |>
  group_by(sub_region) |>
  summarise(
    Count = n_distinct(country_code),
    .groups = "drop"
  ) |>
  arrange(desc(Count))

kable(
  sub_region_counts,
  col.names = c("Sub-region", "Count"),
  caption = "Number of Countries by Sub-region"
)
```

Table 2: Number of Countries by Sub-region

Sub-region	Count
Sub-Saharan Africa	48
Latin America and the Caribbean	33
Western Asia	16
Southern Europe	13
South-eastern Asia	11
Eastern Europe	10
Northern Europe	10
Southern Asia	9
Western Europe	7
Northern Africa	6
Central Asia	5
Eastern Asia	5
Melanesia	4
Micronesia	3
Australia and New Zealand	2
Northern America	2
Polynesia	2

Data visualization using ggplot

```r
ggplot(
  study_regions,
  aes(
    x = fct_infreq(sub_region),
```

```
    fill = region
  )
) +
  geom_bar() +
  coord_flip() +
  scale_x_discrete(limits = rev) +
  labs(
    title = "Number of Countries by Sub-region",
    x = "Sub-region",
    y = "Number of Countries",
    fill = "Region"
  ) +
  theme_minimal()
```